



Towards a new generic framework for citation network generation and analysis in the humanities

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Abstract

Citation network analysis is one of the most developed techniques in science mapping. Various types of citation analysis have been proposed in the literature such as direct citation, co-citation and bibliographic coupling. Networks based on these citation analysis types have been used to reveal discipline structure, central players and emerging trends in STEM and social sciences. However, in the humanities large-scale citation network analysis is still underexplored, mainly due to the lack of comprehensive data sources and varying publication practices. In this paper, we investigate the unique characteristics and needs of citation analysis in the historical humanities and propose a generic framework for systematic generation and analysis of different types of citation networks based on several variables and the essential distinction between citations of primary and secondary sources. The proposed methodology was applied to a corpus of over 15,000 (both primary and secondary) books related to the research field of ancient Mediterranean religions. The obtained results show that in order to gain a deep and comprehensive understanding of a discipline's structure it is beneficial to compare and combine the findings of several types of networks rather than focus on a single network analysis. Comparative community analysis of the networks reveals the existence of a disciplinary core only in the primary literature corpus and a hierarchical sub-discipline structure of the examined research field.

Keywords Citation analysis · Network analysis · Humanities · Primary sources · Community analysis · Betweenness centrality · Digital humanities science mapping

Introduction

Citation network analysis is one of the most prominent tools in science mapping based on large databases of scientific documents (Petrovich, 2020; Petrovich & Tolusso, 2019; Waltman & van Eck, 2014). It aims to show the aggregated relationships between publications

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and authors, that may reflect topics, sub-fields, communities and emerging trends of a given discipline and their inter-connections, or global maps of cross-disciplinary relationships for multi-disciplinary bibliometric corpora (Geffet & Feitelson, 2001; Waltman & van Eck, 2012; Chen, 2017; Van Raan, 2019; Thijs, 2019).

Numerous studies on citation network analysis in various scientific disciplines have been conducted through the past decades (Borgman & Rice, 1992; Chen, 2017; Chi, 2012; Huang et al., 2022). Three different types of citation relationships are used and compared in the literature—direct citation (linking a work to another work it cites), co-citation (linking two works which are both cited in another work) and bibliographic coupling (linking two works which cite a third work) (Small, 1973; Boyack & Klavans, 2010; Petrovich, 2020). Recently, a new type was introduced, a triangular citation relationship, that combines the above three types (Liu et al., 2021). The different networks naturally yield different types of insights and are applicable to different datasets and research objectives (Boyack & Klavans, 2010; Liu et al., 2021).

As opposed to STEM and social sciences, citation analysis has rarely been performed in the humanities and historical disciplines (Ardanuy et al., 2009; Franssen & Wouters, 2019; White, 2015). Challenges include varying publication practices, the importance of books and monographs in the humanities, the lack of comprehensive data sources and tools like the SCI (Science Citation Index) and SSCI (Social Science Citation Index), and the partial coverage of data in databases (Ardanuy et al., 2009; Franssen & Wouters, 2019; Sooryamoorthy, 2020; White, 2015).

As opposed to sciences, where the research materials are the physical world itself or theories about it, in the humanities research centers on texts, and in history—on texts from the period being studied, known as *primary texts* (henceforth: “primary works”). Therefore, publications in historical fields typically include references to both *secondary works* (modern research on the subject) and *primary works* being studied (Hellqvist, 2010). However, a perspective which is seldom recognized in the analysis of citation in the historical humanities is the uniqueness of primary work citation. This unique characteristic of the historical humanities produces different types of citation-based relationships compared to other disciplines. We argue that a citation of one secondary work by another secondary work is essentially different from a citation of a primary work by a secondary work, which is in turn different from a citation of a primary work by another primary work, thus leading to different types of citation networks. Citing a primary work is a different type of intellectual activity from citing a secondary work: primary works are the object of study, while secondary works facilitate research and give it its scholarly and academic context. These citation networks should therefore be constructed and investigated separately. We note that the source type (primary or secondary) is not an intrinsic property of the cited documents, but rather depends on the discipline and bibliographic corpus. Thus, the same document can represent a primary source for a certain area and a secondary source for another. For instance, Wittgenstein’s *Philosophical Investigations*: for a historian of twentieth century philosophy, they represent a primary source, whereas, for a philosopher of language, they represent a secondary source.

There were only few recent studies on science mapping based on citation analysis in the historical humanities. These studies either did not systematically distinguish between different types of works (primary and secondary) and citation networks derived from them or focused on a certain type of citation relationship without considering the others (Romanello, 2016).

The goal of this study is to develop a generic analytic framework for science mapping in the historical humanities based on various types of citation networks. The network

generation framework comprises a few generic variables, such as the type of works used (primary, secondary, both), the type of the network (homogenic, heterogenic), citation relationship (direct, indirect) and reference specification level (title level, page level, reference level). Various combinations of the values of these variables allow to systematically define different types of citation networks for a variety of disciplines in the historical humanities. We specifically focus on the new type of networks generated by the framework—*heterogenic citation networks*. These networks are computed based on both secondary and primary corpora and cross-corpora citations, such as the networks of direct citations of primary works by secondary literature, the networks of primary works co-cited in the same secondary books, the networks of secondary works citing the same primary works. These networks are comparatively investigated through various measures (e.g. centrality and density) and community-based analysis, while discussing their relative advantages and challenges.

This study poses the following research questions:

- (1) What are the differences in expressivity between citation networks constructed by the developed network generation framework and specifically between various heterogenic networks?
- (2) What type of citation network best reflects the core and frontiers of a discipline in the historical humanities?
- (3) What type of network best reflects the communities of a discipline in the historical humanities and their inter-relationships?

To assess the effectiveness of the proposed framework, it was applied to a corpus of academic books on ancient Mediterranean religions. However, the findings are relevant to any historical corpus using primary and secondary works as long as the citations are extractable.

Related work

As mentioned above, existing bibliometric studies in the humanities are relatively rare. They mainly focus on several topics: (i) Comparison of the humanities to other disciplines as regards timeframes for growth, internationality and interconnectivity (Ardanuy et al., 2009); (ii) the (potentially problematic) role of bibliometrics in performance evaluation in the humanities (Melchiorssen, 2019; Nederhof, 2006); (iii) mapping the cognitive and social structure of the humanities, for example, understanding the relationships of various disciplines and sub-disciplines as regards institutional frameworks and personal relations (Chi & Young, 2013; Leydesdorff et al., 2011; Petrovich & Tolluso, 2019). Other studies concern the roles of different types or practices of publication, such as journals vs. monographs (Colavizza, 2018a, 2018b; Franssen & Wouters, 2019; Knievel & Kellsey, 2005), and the comparison of various metrics between researchers in different countries and disciplines (Petr et al., 2021).

Nevertheless, as Colavizza (2017) remarks, “mapping the use of primary sources... has barely started to be explored”, while Must (2012) considered the lack of citation analysis of archival sources as one of the problems of citation analysis in the humanities. Network analysis has been performed on historical primary texts, though such networks have more often been constructed by social links (e.g., letter senders and receivers) rather than citations

(Alexander & Danowski, 1990; Anderson, 2017; Gingras, 2010; Giuffrida, 2014; Lemerrier, 2005; Waxman, 2021; Zhitomirsky-Geffet & Prebor, 2019; Zhitomirsky-Geffet et al., 2020). It has been performed only rarely and at a small scale for heterogenic citations [e.g., the study by Murai and Tokosumi (2006) on co-citation networks of New Testament citations in the works of five Christian theologians]. Another aspect of the humanities, which has barely started to be explored, is mapping the direct citations of primary works in secondary literature, such as (Romanello, 2016) considering the domain of Classics. Recent work by Colavizza (2017, 2018a, 2018b) aimed to study the role of core sources in citation networks in historical research. He used a database of secondary and primary publications on the history of Venice, and showed that monographs create a global network, while journal articles more often belong to smaller local clusters.

Through science mapping techniques based on citation network analysis some new insights can be gained into the structure of the knowledge domain. Thus, previous studies show that while bibliographic coupling network is static, the co-citation network is dynamic and changes over time. As a result, co-citation analysis can better reflect the emerging trends and evolution of the scientific domain (Liu et al., 2021). On the other hand, Newman (2014) argues that the strong co-citation relationship is limited to the highly cited sources, while the coupling relationship is an indicator of more uniform distributions between the sources. They found that the direct citation network can detect emerging research fronts better than the other two analysis types. Boyack and Klavans (2010) showed that the different network types yield completely different clustering results. They analyzed the research front clustering of a large biomedical database and found that bibliographic coupling is better than co-citation analysis and direct citation was the worst. In co-citation analysis, publications are typically seen as a unit—i.e., even if an article is several pages long, each article is seen as one node and there is no differentiation into pages. This is also technically simpler since citations are usually of a whole article. However, in the past decade there has been research into proximity co-citation, i.e., considering the precise placement of each citation in an article, at different levels—same sentence, paragraph or section, and giving higher weights to links between authors and works which are proximate to each other (Callahan et al., 2010; Kim et al., 2016; Liu & Chen, 2012; Tang et al., 2017). These studies have shown that proximity co-citation performs better than traditional co-citation in identifying the core areas of a network while preserving its structure.

A similar systematic analysis of advantages and disadvantages of various citation network types is still needed for the humanities. Must (2012) suggests the following added values that humanities citation databases and their large-scale analysis could provide: sustainability and re-actualization of past studies; tracing the formation of schools and international co-operation communities; facilitating the connection and reuse of sources that were traditionally accessible to and studied only by a limited number of individuals; and exposing a variety of sources to large communities of scholars. In this study, we propose a new methodology that allows to systematically construct and comparatively analyze various citation networks and their potential contributions and challenges for the humanities disciplines' mapping and research.

Methodology

Citation network types

In this study we developed a novel analytic framework for science mapping in the historical humanities based on various types of citation networks. The framework presents

a generic mechanism for citation network generation based on the following variables that determine the type of the network built:

- (1) Source type—primary (historical texts) and secondary (modern research literature) works;
- (2) Citation type—homogenic (within-source-type) citations and heterogenic (cross-source-type) citations;
- (3) Relationship type—direct citation and indirect citation, co-citation and bibliographic coupling;
- (4) Reference specification level—citations at title/book level, or precise page/reference level citations.

Thus, the proposed framework allows to systematically extend the number of networks beyond the three classic network types (direct citation, co-citation, and coupling). It is assumed that some of these will be more useful than others in different scenarios. Providing a wide range of possible options will allow researchers to pinpoint the advantages and problems of each approach. Table 1 details nine network types constructed by variables 1, 2 and 3:

Any of the nine types presented in Table 1 above can be used to construct a network of citations from any historical scholarly corpus, provided the citations can be located and digitized. Homogenic secondary citation analysis (A4–A6) is the most common in the literature. It provides information on the changing landscape of modern historical research, on its communities, frontiers and central players. Homogenic primary citation analysis (A1–A3) provides insights on the textual interactions—and, to a more limited degree, the real-world interactions—of people in the historical period studied. It can pinpoint which texts or people were the most central, or how information travelled in the network at the time.

In this article, however, we focus on heterogenic citation analysis (A7–A9) which is understudied in previous research. Heterogenic direct citation analysis (A7) as well as heterogenic bibliographic coupling analysis (A9) provide information on the scholarly field rather than on the historical period itself. However, they do so based on connections to the research material rather than on connections between contemporary scholars. In A8 the co-citation network is of primary works, clustered according to how they are studied in modern scholarship. The network thus does not give direct information on the historical connections between the primary works, but provides an overview of how modern scholarship perceives, organizes and deploys the primary works: which texts are considered most relevant to each other and should be studied together, and which are considered totally distant from each other.

Thus, heterogenic citation analysis has the potential to produce a different network than homogenic citation analysis, especially in cases where scholars cite similar primary works but not each other—a network representing scholars who are talking about the same thing, even though they are not talking to each other (or, at least, there is no clear evidence that they were talking with each other). Scenarios where a heterogenic network would be significantly different from a homogenic secondary one include:

1. Scholars writing in different languages who may not cite each other, but are studying the same texts.

Table 1 General network types for corpus citation analysis in the humanities (A1–A9)

Corpus division	Homogenic primary	Homogenic secondary	Heterogenic primary and secondary
Direct citation	A1. A primary work citing another primary work	A4. A secondary work citing another secondary work	A7. A secondary work citing a primary work
Co-citation	A2. Primary works co-cited with other primary works in a primary work	A5. Secondary works co-cited with other secondary works in a secondary work	A8. Primary works co-cited with other primary works in a secondary work
Bibliographic coupling	A3. Various primary works citing another primary work	A6. Various secondary works citing another secondary work	A9. Various secondary works citing a primary work

- Scholars working on the same language or period but coming from different scholarly traditions and thus not citing each other. This point is important since disciplinary compartmentalization (e.g., of Classics vs. Theology) may create a different view of scholarship on the subject than that reflected when looking at the works scholars actually study.
- Groups of scholars studying the same texts over the centuries could be better captured by heterogenic networks, as many of these texts have been studied already for hundreds of years.

Table 1 uses and assigns different values to the source type, network type and citation relationship type variables. To consider the specification level variable, each of these nine network types can be used to generate several networks at different specificity levels in primary and secondary works (see Table 2 below for the examples of such networks). In addition, the proposed framework may be expanded with more values (i.e., lower specification levels, such as author, institution and journal level citations) and even new variables as required for the domain of study, and generate more types of citation networks.

Corpus construction

To assess the applicability of the proposed methodology, in this study, primary work indices extracted from scholarly books on ancient Mediterranean religion and culture were collected and analyzed. In practice, this included 364 secondary books in the disciplines of ancient history, classics, ancient philosophy, ancient law, New Testament and patristic studies, Biblical studies, Jewish 2nd Temple literature, and Rabbinics; the majority of the books could be found on the presses' lists by these subjects.

The corpus definition can have a significant impact on the results of citation analysis. This is certainly true in the case of ancient history, which overlaps to various degrees with other disciplines such as linguistics, philosophy, theology, law, and archaeology. Furthermore, while citation analysis of journal articles can make use of the journal framework to determine the discipline of the article, books are not usually published in a specific disciplinary setting. Indeed, in publishers' websites one book may appear under several subjects, demonstrating this ambiguity. Rather than limit the corpus arbitrarily, it was decided to use technical criteria: all books in several leading university and commercial presses

(Oxford University Press, Cambridge University Press, Brill, De Gruyter), which included a primary work index, fully accessible online, and listing more than 100 references to works written in the Mediterranean area between the ninth century BCE and the seventh century CE.

Harvesting the primary work index in secondary works (as opposed to the full text of the secondary work itself) facilitates locating and identifying the references, while maintaining high resolution concerning the precise placement of the reference in the book. A primary work index typically includes a list of primary work authors, primary work title, references to a specific part of the primary work (e.g., chapter and verse references), and the page numbers of the secondary work discussing these references (see Fig. 1). The indices were downloaded from online repositories and parsed using dedicated code which cleaned and unified them, and applied a database of ancient titles, authors and reference styles to identify the works' authors and titles. We constructed this database based on recognized resources on the field, such as the TLG canon of Greek Authors and Works, abbreviation tables from various Greek and Latin dictionaries, and the Classical Works Knowledge Base, but added many more items and expanded also to titles of Coptic, Hebrew, Syriac and other ancient works. Also, ranges of references and page numbers were expanded. The files were then examined by two experts to remove errors or cases in which the database was insufficient.

The result of this process were files of about 3500 rows per book, for a total of 1,272,143 rows, each including the number of the primary work, a precise reference such as “12.7” to denote chapter 12 verse 7, the secondary work from which it came and the page number in that secondary work where the reference appeared. 14,710 unique primary works were referenced in these files.

Corpus characteristics

Most of the books included were published after the year 2000, as only few books published before this date were digitized to date (see Fig. 2).

To give a general sense of the subject-area of the corpus, we tagged the books with the scholarly field we considered most salient. Some books were assigned two labels: “Judaism and Christianity” and “Greek and Latin”; in these cases, the books explicitly discuss both areas. “Cross-category” refers to books which focus on more than two fields, e.g., on antiquity in general, or when it was difficult to provide a more precise designation. As can be seen in Fig. 3, a wide variety of sub-disciplines of ancient history is represented in the corpus. Notable exceptions are art history and archaeology, since the corpus is text-based and includes only few references to artifacts. Furthermore, the periodization used generally excludes texts and studies of the Ancient Near East. This wide range of topics allows us to investigate the links between ancient texts studied in various sub-disciplines, and to analyze the degree of compartmentalization of sub-disciplines. Some books, though meeting the criteria, were not included for technical reasons, such as low image quality or a highly complex or idiosyncratic index structure which made parsing difficult.

Experimental setting

Based on our corpus and the considerations described, we built 12 networks (4 possible combinations of general/specific reference handling, for each of the heterogenic networks

Table 2 Twelve heterogenic citation networks, taking into consideration direct citation/co-citation/bibliographic coupling, and title/specific citation levels

	All Title Both primary and secondary at title level	Secondary specific Primary works at title level, sec- ondary works at page level and primary works at title level	Primary specific Primary works at reference level, secondary works at title level and primary works at reference level	All Specific Both primary and secondary works at reference/page level
Direct citation: Primary works cited by secondary works. Both primary and secondary works explicit as nodes	B1 Nodes: 15,074 Edges: 56,125 AD: 3.70 (SD: ± 15.50, 420%) GC: 100%	B2 Nodes: 96,174 Edges: 260,595 AD: 2.70 (SD: ± 19.5, 707%) GC: 98.60%	B3 Nodes: 475,140 Edges: 744,469 AD: 1.56 (SD: ± 37.8, 2417%) GC: 99.90%	B4 Nodes: 556,240 Edges: 1,298,886 AD: 2.33 (SD: *) GC: *
Co-citation: Primary works explicit as nodes. Secondary works implicit and aggregated as edges weight	B5 Nodes: 14,710 Edges: 4,123,268 AD: 560.6 (SD: ± 2511, 448%) GC: 100%	B6 Nodes: 14,208 Edges: 264,088 AD: 35.90 (SD ± 98.3, 274%) GC: 98.50%	B7 Nodes: ~ 450,000 Edges: ~ 562,500,000 AD: ~ 2500 (SD: *) GC: *	B8 Nodes: 471,947 Edges: 68,631,007 AD: 290.80 (SD: *) GC: *
Bibliographic coupling: Second- ary works explicit as nodes. Primary works implicit and aggregated as edge weights	B9 Nodes: 364 Edges: 58,127 AD: 319.30 (SD: ± 50.50) GC: 100%	B10 Nodes: 81,033 Edges: 63,482,278 AD: 1.566 (SD: *) GC: *	B11 Nodes: 363 Edges: 29,900 AD: 164 (SD: ± 62.2, 37.90%) GC: 100%	B12 Nodes: 74,936 Edges: 2,332,466 AD: 62.20 (SD: ± 119.50, 192%) GC: 96.40%

GC giant component, AD average degree, SD standard deviation

*Currently not analyzed further due to hardware constraints

New Testament I Cor.		Secondary work number	Ancient Work number	Reference	Page number
3:16	152-153	1	8056	3.16	152
5:11-12	151	1	8056	3.16	153
10:11-12	90	1	8056	5.11	151
		1	8056	5.12	151
		1	8056	10.11	90
		1	8056	10.12	90
		1	2309	1.23	51
		1	2309	6.5	233
		1	2309	6.5	234
		1	2309	6.6	233
		1	2309	6.6	234

Fig. 1 Index locorum encoding example

of direct citation, bibliographic coupling and co-citation), and labeled them as B1-B12. Standard measures' scores are provided for each of the networks in Table 2.

Network selection

Based on the measures above, we selected five networks for closer inspection, one from the direct citation type, and two each from bibliographic coupling and co-citation (all bold in Table 2):

- B1: direct citation at title level.

In this directed network, every primary and secondary work was represented as a node, while a pair of secondary and primary works (where the former cites the latter) was connected by a directed edge. The average outdegree of a node (secondary work) is computed as an average number of edges linking it to different primary works. The average indegree is an average number of different secondary works citing (linked to) a node (a primary work).

- B5: co-citation at title level.

In this network, the nodes represent the primary works, while the edges link pairs of primary works that are jointly cited by at least one secondary work. The average degree is calculated as an average number of different primary works co-cited with a given primary work.

- B6: co-citation with secondary works at page level and primary works at title level.

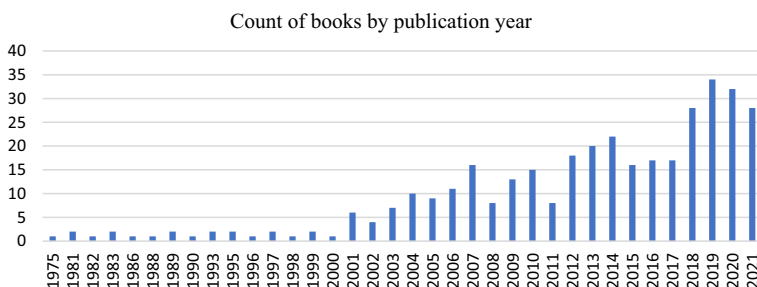


Fig. 2 Book distribution by publication year

In this network, the nodes represent the primary works, while the edges link pairs of primary works that are jointly cited by at least one secondary work on the same page. The average degree is an average number of different primary works co-cited with a primary work on the same page.

- B9: bibliographic coupling at title level.

In this network, the nodes represent the secondary works, while the edges link pairs of secondary books citing at least one same primary work. The average degree is the average number of different secondary works citing the same primary work as a given secondary work.

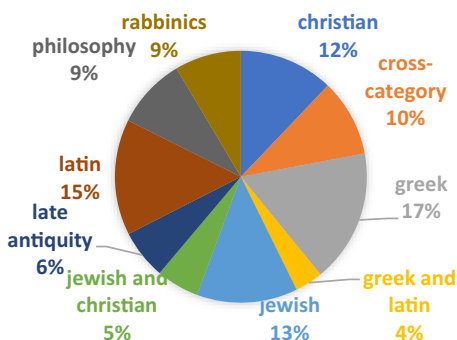
- B11: bibliographic coupling with primary works at reference level and secondary works at title level.

In this network, the nodes represent the secondary works, while the edges link pairs of secondary works citing the same reference in at least one same primary work. The average degree is the average number of different secondary works citing the same reference in a primary work as a given secondary work.

The rationale behind this selection was as follows. As can be observed from Table 2, direct citation, expressing both primary and secondary works explicitly and producing a directed graph from secondary to primary works, has a combination of very low average degree (AD) with a high standard deviation (SD). Hence, it has a small number of edges for each node producing a sparse graph, but some of these edges are weighted much more heavily than others. This makes it generally less useful for bibliometric analysis. We therefore decided to examine further only the smallest network of this type, B1.

In our corpus, the index data easily allows greater specification both for primary and secondary works, since they list not only ancient titles but also references (e.g., chapter/verse or line number of the ancient work), as well as the page on which the primary work reference is cited, thus we can construct networks which utilize this information (see Fig. 1). We hypothesize that two primary works cited together on one page would be more similar to each other, and thus the network formed (B6) would be more significant, than a network based on a co-citation anywhere in the book (title level co-citation) (B5). The question of co-citation of specific ancient references rather than works at title level was also interesting. However, this produced a very large network (B8) with more than 500 million edges, which was difficult not only to plot but also to analyze using standard tools. This network type is thus more relevant for smaller corpora.

Fig. 3 Field breakdown



Similarly, we selected two networks of bibliographic coupling. Here the question of which page on a secondary work is most similar to other pages seemed irrelevant, but a network linking secondary works at title level through the ancient works at reference level (B11) requires comparison with linking through the ancient works at title level (B9).

Network analysis

To demonstrate how the proposed framework can be applied to gain insights on the field of study, the following types of analysis were employed: (1) characterization of the discipline's central players, core and frontiers, mainly based on centrality measures, and (2) mapping the discipline's structure by examining automatically generated communities, relationships between them, and their correspondence to sub-disciplines.

For the first type of analysis, the following steps were performed for each of the five selected networks:

1. Calculation of the distribution of books by relative centrality degrees to identify central players, and core/periphery existence in the corpus.
2. Examination of the top-50 nodes with highest centrality degrees.

Then, to investigate the internal structure of the networks and inter-community relationships, we performed the following:

1. Partitioned the network to communities using the Louvain algorithm (Blondel et al., 2008), available through a Gephi plugin and through Python Community library (<https://github.com/taynaud/python-louvain>). The resolution for partitioning was kept at the default of 1.
2. Manually fitted the communities found by the algorithm to a sub-discipline when possible (e.g., Greek literature, Latin literature, philosophy, early Christianity, Rabbinics, 2nd Temple Judaism, late ancient Christianity). These communities are generally similar for both primary and secondary works, since the scholarly communities are typically organized according to the primary works they study. However, there are many overlaps and ambiguities, both as concerns primary works (e.g., 2nd Temple Judaism, which is studied by scholars of Judaism and Christianity, or late ancient philosophy, which is studied by scholars of Christianity, philosophy and Classics), and as concerns secondary works (e.g., collections of essays on religion in several cultures of the ancient world). Furthermore, though the communities detected by the algorithm mostly corresponded to sub-disciplinary groups, there were always outliers, some of which are discussed below.
3. Calculated the weights of aggregated edges (by the number of citation links in the network) inside and between each of the communities and plotted the data on a chord diagram, to understand the relationships between the communities.

Results

Network characterization, central players and core

B1 is a directed bi-partite network, with edges leading from secondary to primary works only. As opposed to the other networks, B1 does not show to what extent different works are cited together or cite the same works, but simply which secondary works cite which primary works. It is therefore supposed to be best suited for identifying the works cited most often in the literature, which can be considered the core primary works. In Table 2 above, we provide information on the network total average degrees. However, since this is a directed network, its indegree scores differ from outdegree scores, as shown in Table 3 below.

We note that in this stage of the analysis, edges in the networks were unweighted, as using weighted edges for the core literature identification and analysis led to heavily skewed results in some cases. For example, large works which are usually cited by line number (rather than chapter) contain a very large number of references and co-references. Using weighted edges therefore gave these works a strong advantage, even when they were not otherwise so central in the network.

As shown in Fig. 4a and b, B1's outdegrees are about 40 times higher than indegrees, since there are 40 times as many primary as secondary works in the corpus. It is notable that the deviation for indegrees was many times higher than in outdegrees, showing that the variation in importance among primary works as expressed in number of citations was much greater than the variation in the number of references in secondary works. A few secondary works cited as many as 500–800 primary work titles, but the majority cited between 80 and 300 primary work titles. A tiny minority of primary works (145, 1% of the total of works in the database), apparently considered most worthy of study, were cited by as many as 60–180 secondary works, the vast majority (81%) was cited by 1 or 2 secondary works, and the remaining 18% were middle-level works (in terms of citations), which were cited by between 3 and 59 secondary works.

B9 network (bibliographic coupling at title level) shows which secondary works cite the same primary works in common with other secondary works. As opposed to the direct citation network, here citing many and varied works is not the only criteria for rising to the top of the list: it is required that the primary works cited will be cited by many other secondary works as well. The network is very dense with an average relative centrality degree of 319.3 and does not seem to be divided into core and periphery (as shown in Fig. 4c). A large majority of secondary works in the corpus (278, 76%) share at least one primary work cited with more than 300 other secondary works. Those at the top of the degree list share at least one title with 99% of the secondary works in the corpus.

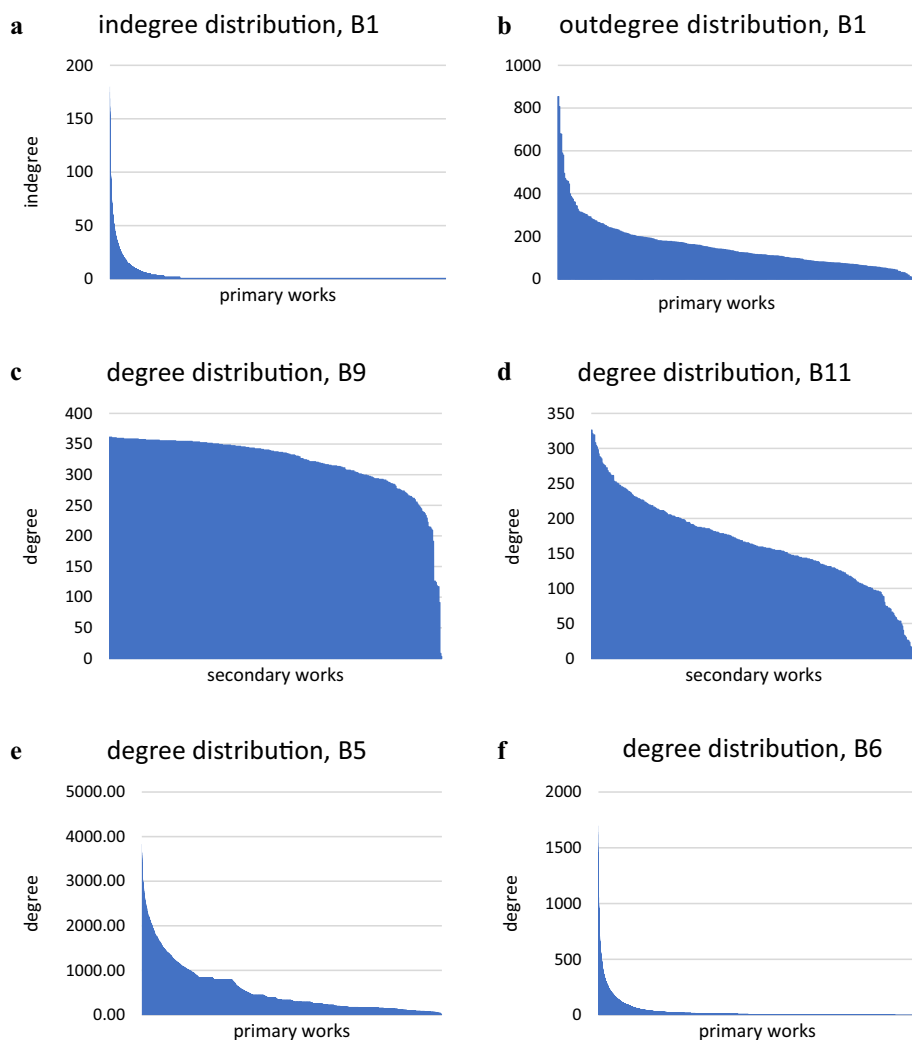
B11 (bibliographic coupling at reference level) represents a subset of B9: they have the same nodes, and all edges appearing in B11 will also appear in B9 (but not vice-versa). It is thus expected that edges which appear in both networks represent a more significant connection than those which appear only in B9. The network is dense with an average degree of 164. The centrality degrees distribution was quite uniform (see Fig. 4d), though less so than in the title level network (B9), since having a shared primary work specific reference is rarer than having a shared title. In this network, secondary works which cite the most primary work references as other secondary works do will have the highest degrees. Only 7 secondary works share at least one primary work reference with more than 300 other secondary works in the network, with a majority (307, 84%) sharing at least one reference

Table 3 Indegree and outdegree centrality scores of B1

Centrality degree	Indegree scores (primary works)	Outdegree scores (secondary works)
Average degree (AD)	1.92 (SD: ± 5.6 , 296%)	77 (SD: ± 56.6 , 73%)

with between 100 and 299 works, and a minority of 48 (13%) sharing with less than 100 secondary works.

B5 network represents co-citation of primary works in secondary works, both at title level. Since the pairing is at title level, any two works cited anywhere in the secondary work are linked with an edge. Distribution of degrees features a clear distinction between


Fig. 4 Centrality degree distributions of books in various networks examined

core and periphery (see Fig. 4e). In other words, as in B1, certain primary works are cited much more often than others. However, the differentiation is much less sharp than in B1, and there is a considerable middle ground. The rank on the degree list depends on how often it is cited in common, but also whether the secondary works it is cited in includes many citations to other primary works. Even an esoteric primary work, cited only once or twice, can have a high degree, as long as it is cited in a secondary book citing many other primary works.

B6 network represents co-citation of primary works on the same page of a secondary works. B6 is a subset of B5: they have the same nodes (primary work titles), but in B6 edges exist only in cases where the two primary work titles are on the same page. Therefore, B6 is less dense than B5 (an average degree of 35.9 vs. 560.6). As shown in Fig. 4f, the centrality degree distribution is highly differentiated, even more so than in B5, with 151 primary works (1.06%) with a degree of more than 500, and 10,370 primary works (72%) with a degree less than 20. The requirement of having a reference on the same page as another eliminates those works which in B5 received a middle score even though they were cited only a few times.

Considering solely the data of the centrality degree distribution at the nodes with highest degrees, each of the five networks examined produce different and complementary types of data. Most importantly, B1 outgoing, B9 and B11 provide information on secondary works, while B1 ingoing, B5 and B6 provide information on primary works. In all cases, the data differs according to the level of analysis, general (at title level) or specific (at reference or page level).

Though there are some overlaps between the sets of data, these are partial. For example, as shown in Table 4, when taking the 68 primary works which together form the 50 top primary works in each of the relevant networks (B1 ingoing, B5, and B6), 35 (51%) are present in all the networks' top-50 lists, a further 12 (18%) in two of the lists, and no less than 21 (31%) appear in only one of the lists.

Therefore, to identify the core primary works of a discipline, it is recommended not to use only one of the networks, but to combine several of them, as an optimized indication of the core works from different perspectives. In our case, for the primary works the combination of strong centrality degrees of B1 ingoing, B5 and B6 indicates not only the most cited (studied) works, but also the most jointly cited with other similar works. Table 5 presents the combined list of the primary works with 10 highest degrees (out of 35 in Table 4, row 1) based on an average of the ranking in all three networks' top-50 works lists:

This list includes works of high diversity as regards period and religion, with three New Testament books, one Hebrew Bible book, three Greek classics, two Latin classics, and one Greek-Jewish work; or, seen by genre, four biblical books, three historiographies, two epics, and one natural history encyclopaedia. The works are all certainly central to the study of ancient history. However, the high representation of biblical material, especially the New Testament, reflects the intense engagement of one sub-discipline, early Christian studies, with this text. This list also excludes many important genres and sub-disciplines: philosophy (three philosophical works are in places 12, 17 and 21); patristics (the first is in place 20); Rabbinics (none in the first 50); epigraphy (39) and papyri collections (62).

In the case of the secondary works, the B1 outgoing, B9 and B11 networks prioritize studies with a comparative or survey approach, like commentaries, since these studies are most likely to include more citations as well as more joint citations with the largest share of other secondary works.

Table 4 The overlap between the top-50 central works in the examined networks

Number of top lists in which the work appears (out of 6)	Number of works	Percent of total (%)
3	35	51
2	12	18
1	21	31

Table 5 The top-10 primary works in descending order of the average ranking of all three networks (B1 ongoing, B5 and B6)

Ancient work	Average ranking
New Testament, Acts	3.00
Homer, Iliad	3.66
Herodotus, Histories	5.00
Hebrew Bible, Genesis	7.00
New Testament, Matthew	7.33
Pliny the Elder, Natural History.	7.66
New Testament, Luke	9.00
Homer, Odyssey.	9.33
Josephus Flavius, Jewish Antiquities	10.00
Diodorus Siculus, Historical Library	11.00

Comparing general level (B9 and B5) to specific level (B11 and B6) networks, there is no clear advantage to the specific level concerning the top lists (i.e. works with the highest centrality degrees in the networks); the top lists of the general level are no less coherent or reasonable than those of the specific level. However, there are significant differences in the middle area of the degree distribution: in the general level networks the middle degrees are much higher than in the specific level networks.

Community structure of the networks

To perform community structure analysis, network graphs were created using Gephi, with layout controlled by the ForceAtlas2 plugin. As opposed to the previous network analysis where edges in the network were unweighted, at the community analysis stage, edges in the networks were weighted according to the number of repetitive citations/co-citations among the same nodes in the network. For example, if in B5 there are three different secondary books citing the same pair of primary books, the edge linking these primary books is assigned a weight of 3. In the larger graphs (B1, B5, B6), the networks were filtered by these edge weights to allow for visualization (see Fig. 5). The communities were created by Louvain algorithm with a resolution of 1, plotted by color in Gephi. As the filtering was of relatively unconnected nodes, it did not have a significant impact on the number of communities or the relationships between them.

Network B1 has a much smaller number of edges per node than either co-citation or bibliographic coupling networks, because primary works are not linked among themselves but only to the secondary works. The high centrality degree deviation in the network can be seen on the graph (Fig. 5), with some nodes much bigger than others. The Louvain algorithm identified 10 communities. However, an examination of the contents of these

communities found that only a few of them contained clear disciplinary groups of titles, while most of them were mixed. This can also be seen graphically in the mixing of edge colors in the graph. We therefore decided not to continue with community analysis of this network where the communities are not well-delineated.

In network B9, at a resolution of 1, the Louvain algorithm produces only two major communities, corresponding to Greek/Latin Classics (green), and Jewish/Christian literature (purple).

As can be observed from Table 6, the density of the Jewish/Christian community in B9 is almost double that of the Greek/Latin community, showing that the reason for the density is not (only) the number of discrete references but also the number of titles cited together. Thus, it appears that the corpus used by scholars of the Jewish/Christian community is less diverse than that of the Greek/Latin community.

Network B11 produces a relatively clear graph with four communities (see Fig. 5), corresponding to scholars working on the fields of Greek and Latin (purple), Jewish studies (orange), Christianity (green), Magic and some cross-category books (light blue). To examine the relationships between and inside the communities, we first found the densities of each community as shown in Table 6.

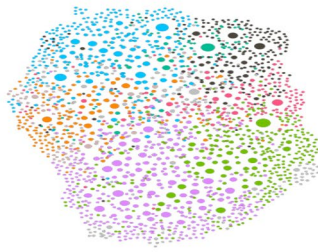
Disregarding the small community (Magic) which is clearly an outlier, the relative densities of each community can be clearly seen. Table 7 indicates that Jewish studies and Christian studies cite the same reference more often, while Greek and Latin (Classics) are more diverse. This generally makes sense, since much of the referencing in Jewish and Christian studies are of the Bible, a single (but large) work. Furthermore, in Classics the works' references are frequently references by line number, leading to a very large number of discrete references, while in Rabbinnics, for example, citations are usually of a folio number containing a large amount of text, leading to a smaller amount of overall references. Therefore, as expected, a network based on work titles rather than discrete references (B9) showed a coarser grained community distinction picture than B11. To investigate the relationships between the different communities, we produced a chord diagram of the cumulative weights of edges within and between each of the communities (Fig. 6). Though the Jewish and Rabbinic disciplines have the highest number of internal edges, they also have a high number of links with Christian/Late Antiquity studies.

As shown in Figs. 5 and 6, four communities were detected with a resolution of 1 in the filtered (edge weights > 5) B5 network. These were identified as: Jewish, Christian, Greek and Latin. The densest community is Jewish literature (Table 8), while the largest and sparsest community is Greek.

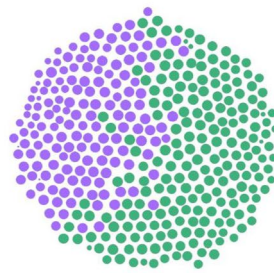
In B6, the algorithm identified a total of 65 communities with a default resolution of 1, but most of these include only one or two texts; filtering these out, 98% of the nodes of the network comprise nine communities (color coded in the graph). These communities clearly correspond to groups of texts which are commonly studied together. Some texts, however, were near the margins of the community, and were, moreover, linked to many other communities; for example, Josephus' *Jewish Antiquities* was linked to all nine communities, while several biblical books and Stabo's *Geography* were linked to eight. The nine communities are presented in Fig. 6 and Table 9.

Community structure detection is one of the most intriguing questions in citation network analysis and science mapping. To this end, the abovementioned results can be combined to provide argumentation for network partitioning into communities. For example, it can be observed from Figs. 5 and 6 that in B9 there are only two large and general communities (Jewish/Christian and Greek/Latin). In B5 each of these were split

B1: direct citation network



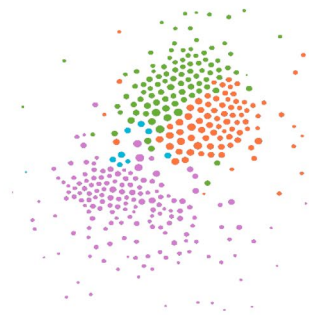
B9: bibliographic coupling at title level



B5: co-citation at title level



B11: bibliographic coupling at ref. level



B6: co-citation at page level



Fig. 5 The five network graphs. Community partitioning was marked by a color scheme, centrality degree by node size. Edges and labels are omitted for clarity. Network B1: 2065 (13.6%) nodes plotted, filtered by edge weight > 200. Network B6: 2175 (15.3%) nodes plotted, filtered by edge weights > 5. Network B5: 1849 (12.5%) nodes, filtered by edge weights > 5

Table 6 B9 community metrics

Community	Total edge weights	Number of nodes	Density
Greek/Latin	187,870	193	10.14
Jewish/Christian	280,831	171	19.32
All network	514,791	364	15.12

into two further categories. In B11 Magic has formed a separate community. Finally, in B6 Rabbinic literature becomes a separate community, as it is only connected (cited/ studied together) with the 2nd Temple Jewish community and has no connections with any other community in the network, while 2nd Temple Jewish is connected to all the other communities in the network. This indicates that Rabbinic literature is characterized differently than 2nd Temple Jewish and others, which supports it being a separate community. On the other hand, in B6 Early Christianity is very strongly connected with 2nd Temple Jewish, and both communities' connections with the rest of the communities are similar, as shown in Fig. 6. This provides an argument for merging these two communities into a single community which is also supported by networks B9 and B5. This integrative analysis of various networks may also lead to the conclusion that there is a hierarchical structure of communities and sub-communities in the examined discipline (as shown in Fig. 7).

Discussion and conclusion

Citation network analysis is one of the most developed techniques in science mapping (Petrovich, 2020). Various citation types have been proposed in the literature such as direct citations, co-citation and bibliographic coupling. Networks based on these citation types have been extensively analyzed to reveal discipline structure, central players and emerging trends in STEM and social sciences (Borgman & Rice, 1992; Chen, 2017; Chi, 2012; Huang et al., 2022). However, in the humanities large-scale citation network analysis is still under-researched mainly due to the lack of comprehensive data sources and varying publication practices (Ardanuy et al., 2009; Franssen & Wouters, 2019; Sooryamoorthy, 2020; White, 2015).

In this paper, we introduce a generic framework for systematic generation and analysis of different types of citation networks in the historical humanities. To this end, we explore the unique characteristics of research literature and citations in the humanities focusing on historical research, and propose to distinguish between secondary works (modern research literature and citations present in it) and primary works (historical books and their citations). Based on this distinction, new citation types can be defined, namely cross-corpora citations, that in contrast to the traditionally studied within-corpus (homogenic) citations lead to the creation of heterogenic citation networks.

Our findings from the analysis of the corpus of over 15,000 books on ancient Mediterranean religions demonstrate that the comparative exploration of heterogenic citation networks of different types and specification levels allows to better capture the structure and knowledge distribution of the discipline, such as: (i) the presence of core only in the primary literature corpus; (ii) the relatively uniform distribution and high connectivity of the

Table 7 B11 community metrics

Community	Total edge weights	Number of nodes	Density
Magic	16,075	10	357.22
Christian	140,115	84	40.19
Jewish	136,966	102	26.59
Greek/Latin	165,609	165	12.24
All network	687,456	364	10.40

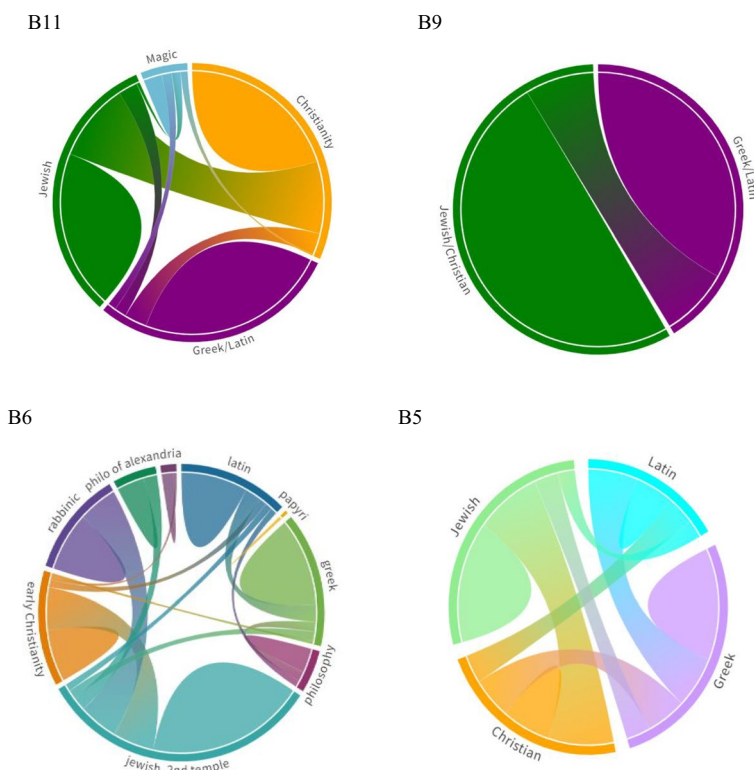


Fig. 6 Chord diagrams of the internal and external community relationships of the networks examined

Table 8 B5 community metrics

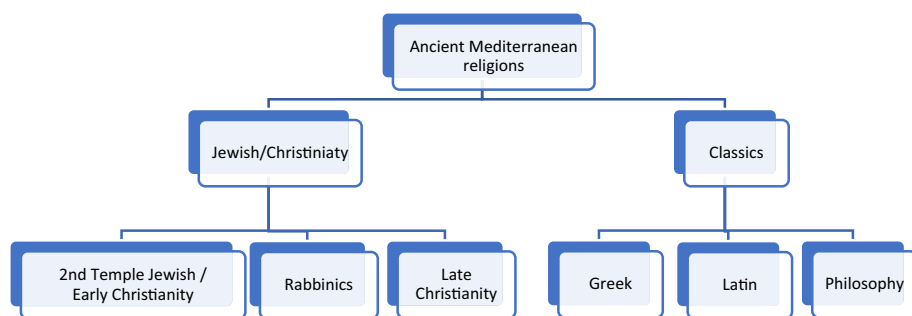
Community	Total weight of edges	Number of nodes	Density
Latin	131,656	354	2.11
Greek	355,611	712	1.40
Christian	163,697	410	1.95
Jewish	386,885	373	5.58
All network	2,535,479	1849	1.48

secondary literature; and (iii) the division into communities and sub-communities emerging from the integration of the results of several types of networks rather than from a single network type.

Clearly, diverse and complementary insights can be gained by looking at different network types. In addition, the study suggests that humanities scholars who adopt the proposed framework will become aware of the various types of citation networks and their generation mechanism, and consequently will be able to make an informed choice of those mostly fitting to their research objectives. The main conclusion is that there is no one best network type for a discipline in the humanities, but rather a combination and mutual

Table 9 Network B6 metrics

Community	Total weight of edges	Number of nodes	Density
Greek literature	34,607	536	0.24
Latin literature	27,156	361	0.42
2nd Temple Jewish	64,340	302	1.42
Early Christianity	20,498	279	0.53
Rabbinics	23,309	237	0.83
Philosophy	9,592	178	0.61
Late antique Christianity	4,788	176	0.31
Papyri	1,184	63	0.61
Philo of Alexandria	11,220	43	12.43
All network	322,380	2175	0.13

**Fig. 7** Community structure of the field of Ancient Mediterranean religions inferred from the integrative network analysis

cross-validation of multiple networks' results may lead to the most precise indications regarding the various disciplines' characteristics and structure. The results of the study also demonstrate that different types of references used in different sub-disciplines (e.g., referencing by line number vs. chapter number) can have an impact on the network shape; and that more specific levels for constructing networks (page and reference vs. title) result in higher differentiation into communities, but also require more elaborate data collection, and, especially with large corpora, higher hardware requirements.

In future research, a systematic comparison of networks with weighted and unweighted edges can be performed. The proposed framework can be extended with more variables (e.g. more proximity and aggregation levels) and refined corpus types (e.g. studying commentaries and surveys as a separate type of corpus, or comparing between citation networks of monographs vs. edited books and journal articles). In addition, more detailed analysis of each network and its communities can raise and answer other questions, for example, identification of specific works at the core and margins of each of the communities, and identification of works bridging communities. As in the top-degree lists of the overall networks, it is expected that there will be overlaps in these areas, but also that each network will provide different perspectives. Furthermore, additional insights may be gained by using innovative network analysis methodologies, such as multilayer networks

(Bianconi, 2018) or network fusion (Baccini et al., 2022), which allow combining different types of interacting networks.

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Declarations

Conflict of interest The authors declare they have no conflict of interest regarding this research.

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